Electricity market and smart meter rollout

Brazilian regulatory framework

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THE BRAZILIAN ELECTRICITY SECTOR

The Brazilian Electricity Sector in NUMBERS

NUMBER OF CONSUMERS
91.6 MILLION
212 MILLION pop.

ACCESS TO ELECTRICITY
99,8%
of the population

CENTRALIZED GENERATION CAPACITY
218 GW

DISTRIBUTED GENERATION CAPACITY
45.7 GW
4 million plants

MAXIMUM LOAD
106.5 GW
(February 2025)

TRANSMISSION
2022: 179.300 km
2027: 216.000 km

ELECTRICITY LOSSES (2024)
14%
TL: 7.4% **NTL: 6.6%**

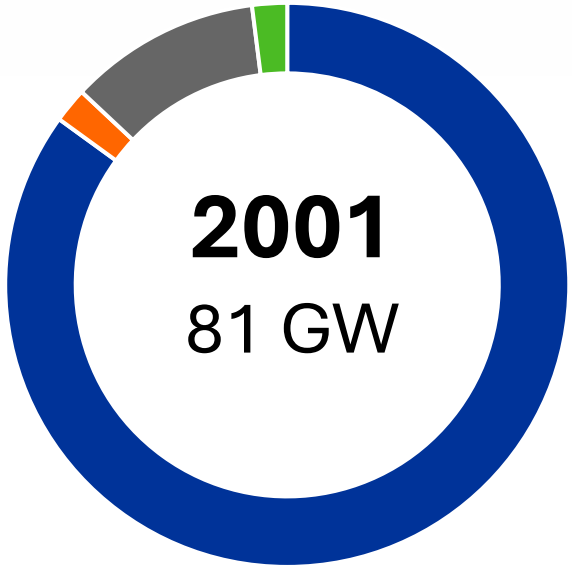
Evolution of the Brazilian Electricity Mix

Nuclear
1,6GW
2%

Biomass
1,6GW
2%

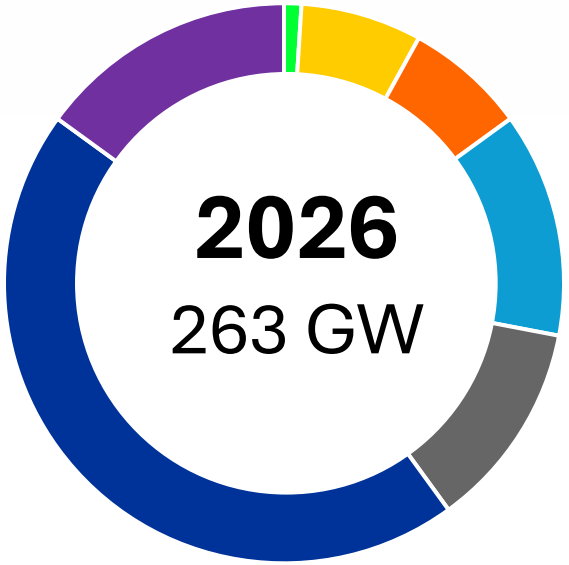
Fossil
8,9GW
11%

Hydro
68,9GW
85%



+225%

182GW



Wind
34.8GW
13.2%

Solar
21.6GW
8,2%

DER
45.7GW
17.4%

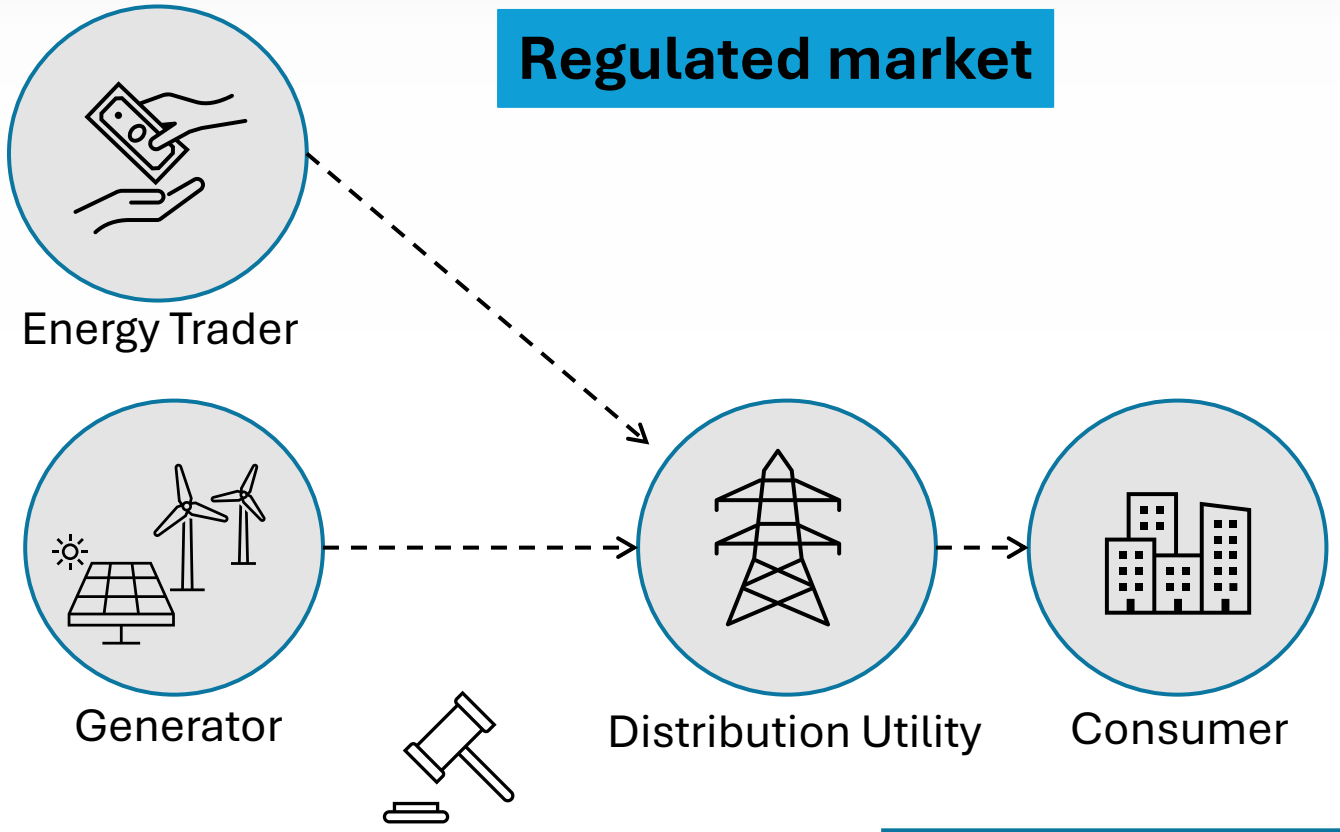
Nuclear
2GW
1%

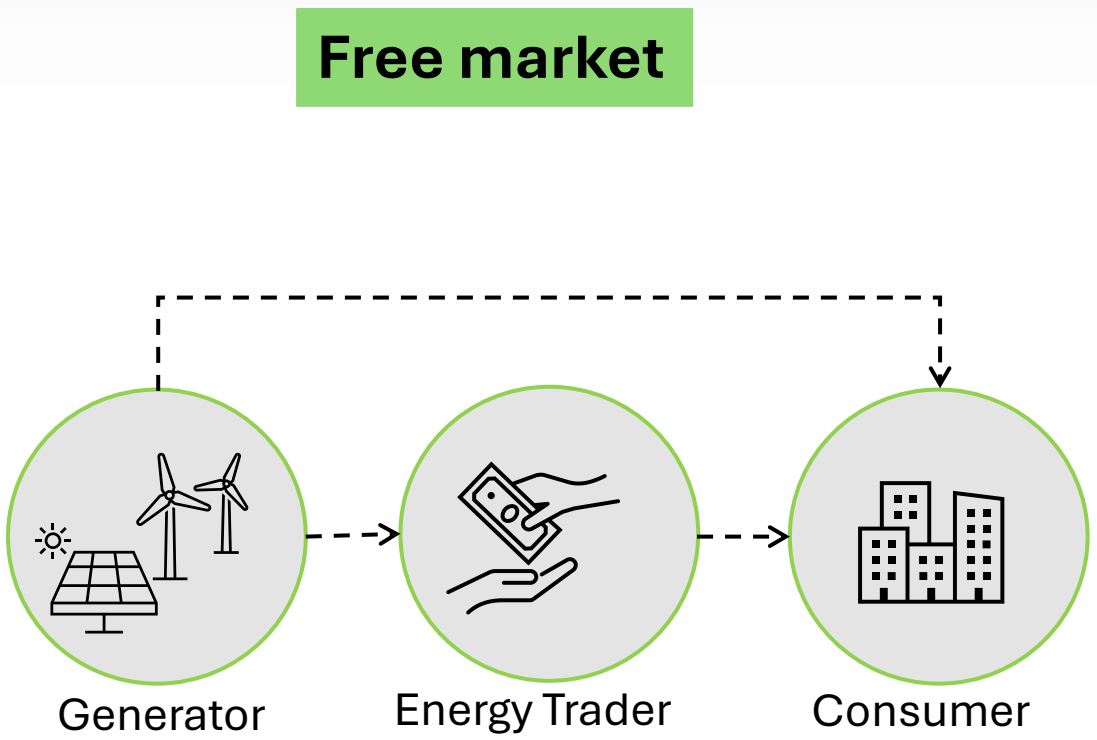
Biomass
18.1GW
6.9%

Fossil
31.4GW
12%

Hydro
110.2GW
41.9%

Brazilian Electricity Market





40,393 MWm
57.6%

29,757 MWm
42.4%

Adapted from CCEE
https://www.ccee.org.br/en/web/guest/mercado/adesao

ELECTRICITY MARKET LIBERALIZATION: PROCESS AND EVOLUTION

The free market opening

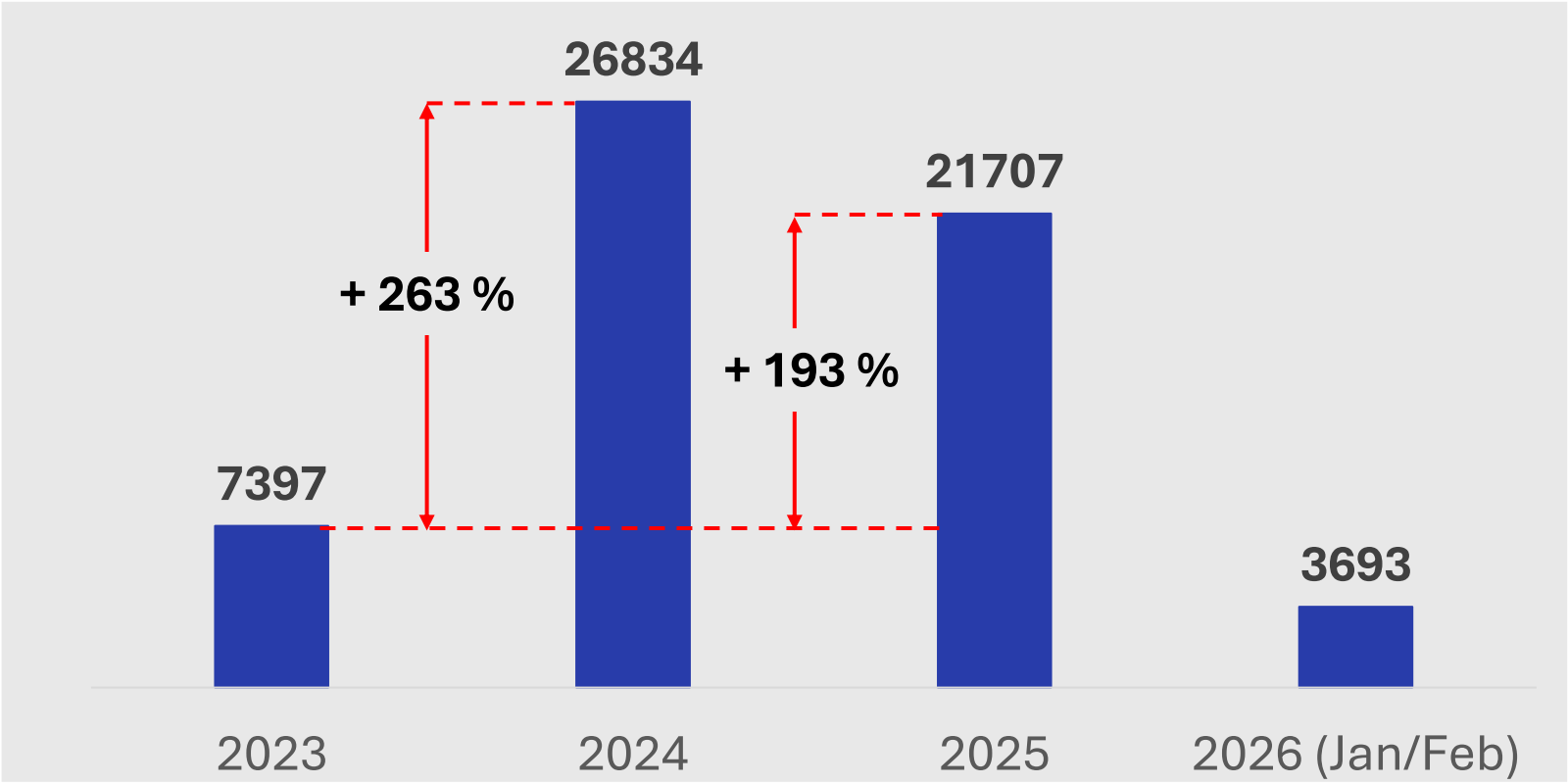
Prior to 2019: only consumers above 3,000 kW were eligible for the free market

We are here!

Ordinance No. 514 Dec. 2018		Ordinance No. 465 Dec. 2019			Ordinance No. 50, Sep. 2022	Law No. 15,269 Nov. 2025	
2019	2020	2021	2022	2023	2024	2027	2028
July	July	January	January	January	January	Nov. 25th	Nov. 25th
≥ 2,500 kW	≥ 2,000 kW	≥ 1,500 kW	≥ 1,000 kW	≥ 500 kW	All medium voltage level	Low voltage commercial industrial	Low voltage residential others

The free market opening

Evolution of migration to the free market



Ordinance No. 50, Sep. 2022

The free market opening

Topics currently under discussion

Switching to the free market and returning to the regulated market: conditions and timeframes

Billing: two separate bills (retailer - distribution utility)?

Electricity metering requirements (hourly-based settlement)

Supplier of last resort: timeframes and conditions

Open Energy: data sharing procedures

Comparative platform for prices and products: transparency

Communication plan

SMART METER ROLLOUT

Smart meter rollout

Main drivers in the Brazilian context

- **Non-technical loss** mitigation
- **Tariff modernization** (TOU, prepayment)
- **Market opening** (hourly-based settlement and opportunities for dynamic contracts)
- **Demand response** in a context of high penetration of variable energy resourses (VERs)
- **Operational cost reduction** (remote reading and remote control)
- **Improved observability** of distribution systems for planning and operational purposes

Smart meter rollout

What we need to regulate

1 Cost-benefit analysis

- **Cost-benefit analysis** to support large-scale smart meter deployment
- **Guidelines** for the analysis
- **Who** is responsible for conducting the analysis (distribution utilities)
- **Procedures** when the cost-benefit analysis **is positive or negative**

2 Technical requirements

- **Definition** of a smart meter
- **Technical requirements**: bidirectional metering, memory capacity, remote communication capabilities, interoperability, measurement granularity and other advanced metering functionalities

Public hearing No 1 of 2026

ANEEL regulation expected by the end of 2026

“As for the future, your task is not to foresee it, but to enable it”

Antoine de Saint Exupéry

ANEEL

Agência Nacional de Energia Elétrica

30 ANOS

Energia que move o futuro